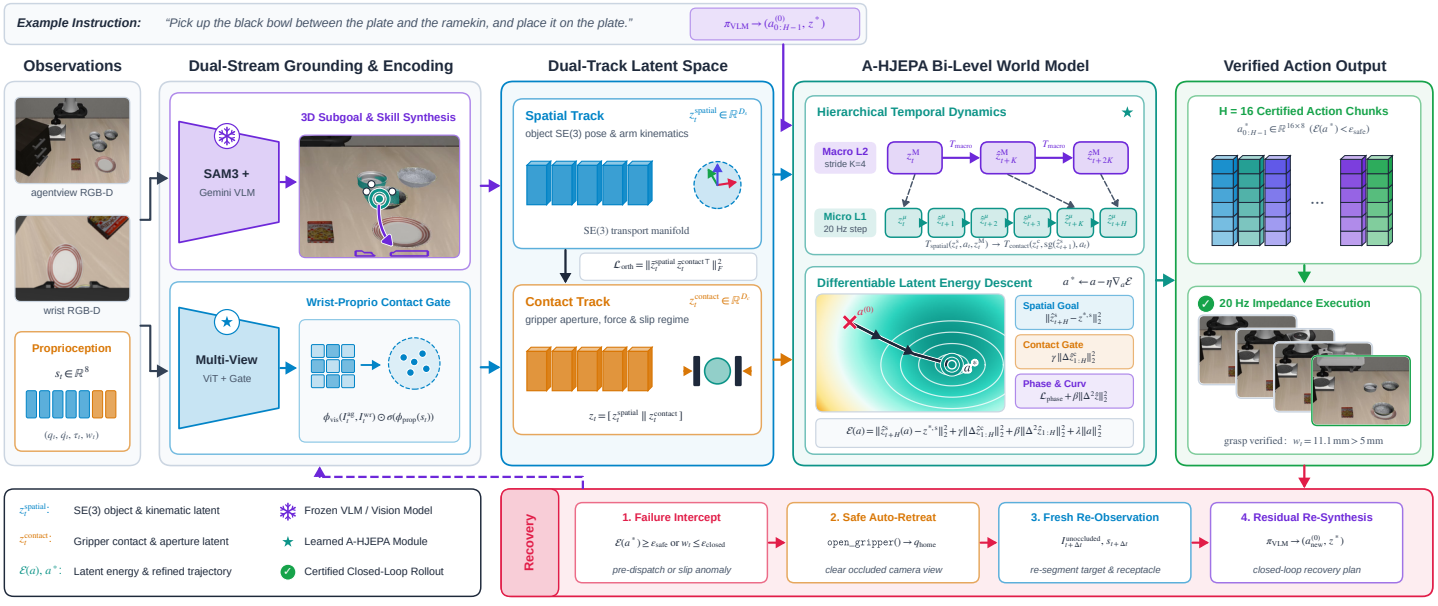

Example Instruction:
"Pick up the black bowl between the plate and the ramekin, and place it on the plate."
$\pi_{\text{VLM}} \to (a^{(0)}_{0:H-1}, z^*)$
Observations
agentview RGB-D
wrist RGB-D
Proprioception
$s_t \in \mathbb{R}^8$
$(q_t, \dot{q}_t, \tau_t, w_t)$
Dual-Stream Grounding & Encoding
SAM3 +
Gemini VLM
3D Subgoal & Skill Synthesis
Multi-View
ViT + Gate
Wrist-Proprio Contact Gate
$\phi_{\text{vis}}(I^{\text{wr}}_t, I^{\text{ag}}_t) \odot \sigma(\phi_{\text{prop}}(s_t))$
Dual-Track Latent Space
Spatial Track
$z^{\text{spatial}}_t \in \mathbb{R}^{D_s}$
object SE(3) pose & arm kinematics
SE(3) transport manifold
$\mathcal{L}_{\text{orth}} = \| z^{\text{spatial}\top}_t z^{\text{contact}}_t \|^2_F$
Contact Track
$z^{\text{contact}}_t \in \mathbb{R}^{D_c}$
gripper aperture, force & slip regime
$z_t = [\, z^{\text{spatial}}_t \,\|\, z^{\text{contact}}_t \,]$
A-HJEPA Bi-Level World Model
Hierarchical Temporal Dynamics
Macro L2
stride K=4
Micro L1
20 Hz step
T_{macro}
$T_{\text{spatial}}(z^{\text{s}}_t, a_t, z^{\text{M}}_t) \to T_{\text{contact}}(z^{\text{c}}_t, s_t(z^{\text{s}}_{t+1}), a_t)$
Differentiable Latent Energy Descent
$a^+ \leftarrow a - \eta \nabla_a \mathcal{E}$
$a^{(0)}$
Spatial Goal
Contact Gate
Phase & Curv
$\mathcal{E}(a) = \| z^{s,+}_{t+H} - z^{s,*}_{t+H}(a) \|^2_2 + \gamma \| \Delta z^{\text{c}}_{1:H} \|^2_2 + \beta \| \Delta^2 z_{1:H} \|^2_2 + \lambda \|a\|^2_2$
Verified Action Output
H = 16 Certified Action Chunks
$a^+_{0:H-1} \in \mathbb{R}^{16\times 8}$ $(\mathcal{E}(a^+) < \varepsilon_{\text{safe}})$
20 Hz Impedance Execution
grasp verified: $w_t = 11.1\,\text{mm} > 5\,\text{mm}$
z^{spatial}_t: SE(3) object & kinematic latent
z^{contact}_t: Gripper contact & aperture latent
$\mathcal{E}(a)$, a^*: Latent energy & refined trajectory
Frozen VLM / Vision Model
Learned A-HJEPA Module
Certified Closed-Loop Rollout
Recovery
1. Failure Intercept
$\mathcal{E}(a^*) \geq \varepsilon_{\text{safe}}$ or $w_t \leq \varepsilon_{\text{closed}}$
pre-dispatch or slip anomaly
2. Safe Auto-Retreat
open_gripper() $\to q_{\text{home}}$
clear occluded camera view
3. Fresh Re-Observation
$I^{\text{reocluded}}_{t+\Delta t}, s_{t+\Delta t}$
re-segment target & receptacle
4. Residual Re-Synthesis
$\pi_{\text{VLM}} \to (a^{(0)}_{\text{new}}, z^*)$
closed-loop recovery plan